Nonlinear Practice 1

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1. A 143 lb, 21-year-old woman is receiving phenytoin for the treatment of seizures. When she was taking a daily oral dose of 300 mg phenytoin, her steady-state plasma concentration was 5.1 mg/L. Because the phenytoin plasma concentration was well below the therapeutic range, the patient's dose was increased to 500 mg phenytoin, which resulted in a steady-state plasma concentration of 20 mg/L.

a (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) a. Calculate the patient's Vmax and KM. What is the KM?

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b (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) b. Recommend a daily dose of phenytoin for this patient to achieve a phenytoin plasma concentration of 15 mg/L at steady state.

c (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) c. Calculate the half-life and the total body clearance when the plasma concentration is 15 mg/L. What is the half-life?

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d (Assume that the absolute bioavailability of phenytoin is 100% and the volume of distribution is 0.7 L/kg.) d. What will be the steady-state concentration that should be achieved if the dose were 450 mg daily?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 5.91 mg/L

WHAT THE STEM GIVES

R1	300 mg	the first daily dose
C_1	5.1 mg/L	steady state on 300 mg
R2	500 mg	the increased daily dose
C_2	20 mg/L	steady state on 500 mg
weight	143 lb	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 300 mg/day, C_1 = 5.1 mg/L, R2 = 500 mg/day, C_2 = 20 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method gives KM first, and Vmax follows from it.

ALGEBRA

$$KM = \frac{500 - 300}{(300/5.1) - (500/20)} = \frac{200}{58.82 - 25} = \frac{200}{33.82}$$

Each dose over its own concentration. The 143 lb weight and the 0.7 L/kg volume play no part in KM; they are needed only for the half-life in part c.

ROUND

KM = 5.91 mg/L

(mg/day) over (mg/day per mg/L) leaves mg/L. Two decimal places carry enough precision into Vmax and the later parts.

> **DOES THE NUMBER MAKE SENSE?**
>
> A two-thirds rise in dose nearly quadrupled the concentration, the mark of saturation; KM is a concentration: 5.91 mg/L.

a

Her key

Answer: 647.8 mg/day

WHAT THE STEM GIVES

KM	5.91 mg/L	from the KM half of part a
R1	300 mg	the dose used
C_1	5.1 mg/L	its own steady state

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax, with one dose and the steady-state concentration that dose produced; the other pair gives the same value.

ALGEBRA

$$\text{Vmax} = \frac{(300\ \text{mg/day})(5.91304 + 5.1)\ \text{mg/L}}{5.1\ \text{mg/L}} = \frac{3303.91}{5.1} = 647.83\ \text{mg/day}$$

KM is carried unrounded, 5.913 mg/L: her 647.83 mg/day comes from it, and KM rounded to 5.91 gives 647.6 mg/day. mg/L cancels and Vmax keeps the units of the dose, mg/day. The 500 mg/day with 20 mg/L pair gives 500 × 25.913 over 20 = 647.83 mg/day as well.

ROUND

$$\text{Vmax} = 647.8\ \text{mg/day}$$

647.83 mg/day to one decimal place. Her sheet keeps 647.83 mg/day and carries it into the dose, half-life, clearance and C_{ss} of parts b to d.

DOES THE NUMBER MAKE SENSE?

Vmax must exceed 500 mg/day, the higher dose that reached a steady state: 647.8 mg/day.

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b

Her key

Answer: 465 mg/day

WHAT THE STEM GIVES

KM	5.91 mg/L	from part a
Vmax	647.83 mg/day	from part a
C_{ss}	15 mg/L	the target

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{\text{Vmax} \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax from part a, with the target of 15 mg/L as C_{ss}, in the nonlinear dose-rate line; no weight is needed because Vmax is in mg/day.

ALGEBRA

$$R = \frac{(647.83\ \text{mg/day})(15\ \text{mg/L})}{(5.91 + 15)\ \text{mg/L}} = \frac{9717.5}{20.91} = 464.7\ \text{mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between 5.1 and 20 mg/L, so the dose lies between the 300 and 500 mg/day she was given.

ROUND

$$R = 465\ \text{mg/day}$$

464.7 mg/day rounded to the nearest 5 mg is 465 mg/day, a dose that can be given; the unrounded 464.7 mg/day is also accepted.

DOES THE NUMBER MAKE SENSE?

15 mg/L lies between 5.1 and 20 mg/L, so the dose lies between 300 and 500 mg/day: 465 mg/day.

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c Her key

Answer: 1.02 days

WHAT THE STEM GIVES

weight	143 lb	÷ 2.2 = 65 kg
V_D	0.7 L/kg	× 65 kg = 45.5 L
C_p	15 mg/L	the stated concentration
KM	5.91 mg/L	from part a
Vmax	647.83 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

UNIT

$$\text{weight} = \frac{143\ \text{lb}}{2.2\ \text{lb/kg}} = 65\ \text{kg};\ V_D = 0.7\ \text{L/kg} \times 65\ \text{kg} = 45.5\ \text{L}$$

The weight is in pounds and the volume per kilogram, so pounds go to kilograms first, then the volume to litres, which the half-life line needs.

SETUP

$$t½ = V_D\left(\frac{0.693}{Vmax}\right)(KM + C_p)$$

For a saturable drug the half-life depends on concentration, so it is calculated at the stated 15 mg/L, not from 0.693 over a fixed k.

ALGEBRA

$$t½ = 45.5\ \text{L} \times \frac{0.693}{647.83\ \text{mg/day}} \times (5.91 + 15)\ \text{mg/L} = 1.018\ \text{days}$$

L × mg/L gives mg, which cancels against mg/day and leaves days. Using 143 as kilograms gives 2.24 days, a unit slip.

ROUND

$$t½ = 1.02\ \text{days}$$

1.018 days to two decimal places. The half-life is longer at higher concentrations because KM + C_p grows while Vmax stays fixed.

DOES THE NUMBER MAKE SENSE?

Vmax is in mg/day, so the half-life comes out in days: 1.018 days, about one day, not a fraction of an hour.

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c Her key

Answer: 30.98 L/day

WHAT THE STEM GIVES

C_p	15 mg/L	added to KM in the denominator
KM	5.91 mg/L	from part a
Vmax	647.83 mg/day	from part a

weight	143 lb	*not needed for clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}$$

For a saturable drug, clearance is calculated at a stated concentration, because C_p sits in the denominator and changes the result.

ALGEBRA

$$Cl_T = \frac{647.83 \text{ mg/day}}{(5.91 + 15) \text{ mg/L}} = \frac{647.83}{20.91} = 30.98 \text{ L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The weight and V_D are not needed for clearance.

ROUND

$$Cl_T = 30.98 \text{ L/day}$$

To two decimals, as her sheet gives it. At 20 mg/L the clearance would be lower, 25.0 L/day, because the concentration in the denominator is larger.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance at 15 mg/L is well below Vmax ÷ KM, because KM + C_p, 20.91 mg/L, is larger than KM alone: 30.98 L/day.

d

Her key

Answer: 13.5 mg/L

WHAT THE STEM GIVES

R	450 mg daily	the new dose rate
KM	5.91 mg/L	from part a
Vmax	647.83 mg/day	from part a

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R \times KM}{Vmax - R}$$

The dose-rate line solved for C_{ss}: multiplying out gives R × KM + R × C_{ss} = Vmax × C_{ss}, so R × KM = C_{ss}(Vmax − R), then divide by (Vmax − R).

ALGEBRA

$$C_{ss} = \frac{(450 \text{ mg/day})(5.913 \text{ mg/L})}{(647.83 - 450) \text{ mg/day}} = \frac{2660.9}{197.83} = 13.45 \text{ mg/L}$$

KM is carried unrounded, 5.913 mg/L; with 5.91 the quotient is 13.44 and would round to 13.4. mg/day cancels and leaves mg/L. The denominator subtracts the dose from Vmax; adding them gives 2.4 mg/L, below even the 5.1 mg/L that 300 mg produced.

ROUND

C_{ss} = 13.5 mg/L

13.45 mg/L to one decimal place is 13.5 mg/L, her sheet's value. It lies between the 5.1 and 20 mg/L of the two measured doses.

DOES THE NUMBER MAKE SENSE?

450 mg/day lies between 300 and 500 mg/day, so C_{ss} lies between 5.1 and 20 mg/L: 13.5 mg/L.
